Homework 1: Preliminaries

The CypherStack Team

Instructions

The purpose of this homework assignment is to review the concepts covered in Lecture 1 and introduce ideas that we did not have the time to cover. In the case that new material appears here that was not covered in class, please make note of any new notation or results, as they will be used in later lectures.

Please attempt each problem. Even if you are not successful, it is important to at least write down your thoughts on the problem and note where you are getting stuck, as well as any points of confusion.

New Material

Definition 1 (Binary Representation)

Let $N \in \mathbb{N}$. Every positive whole number has a **binary representation**, which is of the following form:

$$N = \sum_{i=0}^{k-1} b_i \cdot 2^i$$

where $b_i \in \{0, 1\}$ and $b_{k-1} = 1$.

The **number of bits** in N is equal to $k = \lfloor \log_2(N) \rfloor + 1$. This quantity is denoted by $len(N)$.

Example 2

Let's compute the binary expansion of 13. The easiest way to do this is by repeatedly dividing by 2 and keeping track of the quotient:

Value	Quotient when divided by 2	Remainder
13	7	1
7	3	1
3	2	1
2	1	0

Once we have a quotient of 1, we can stop. Then it is as simple as reading the remainder row from top to bottom! So, the binary representation for 13 is simply $(1110)_2$.

Problem 1 (Binary Representation)

Compute the binary representation of the following numbers:

(a) 121

(b) 63

(c) 24

(d) 73

Compute the decimal representation from the following binary representations:

(e) $(10101)_2$

(f) $(10110)_2$

(g) $(10001001)_2$

(h) $(10111010)_2$

Definition 3 (Big O Notation)

Let $f(x)$ and $g(x)$ be functions on the same domain D. We say that:

$$f(x) = O(g(x)$$

if there exists some positive real number M such that, for all x in the domain D:

$$|f(x)| \leq M \cdot g(x)$$

The above definition is a little technical, but the purpose of it is to show how quickly or slowly certain quantities grow, in terms of common functions. In cryptography, we use this notation to describe how long certain operations might take. For example. suppose we have two values we want to add together: a and b.

To make life a little more simple, we will suppose that a and b are both n bit integers. If we were to attempt to determine the 'cost' of adding these two values together, what exactly does that look like?

Let's look at the binary representation of a and b. Since they are n bit integers, we can represent them as:

$$a = (a_{n-1}a_{n-2}...a_1a_0)_2$$

$$b = (b_{n-1}b_{n-2}...b_1b_0)_2$$

To add them together, we simply need to go 'column by column', adding the bits and keeping track of the carry over value. We can make a table to tabulate the result, then we need only keep track of how many times we perform certain operations:

Addition	Bit	Carry Over	Operation Count
$a_0 + b_0$	c_0	d_0	1 addition
$d_0 + a_1 + b_1$	c_1	d_1	2 additions
$d_1 + a_2 + b_2$	c_2	d_2	2 additions
$\vdots$	$\vdots$	$\vdots$	$\vdots$
$d_{n-2} + a_{n-1} + b_{n-1}$	c_{n-1}	d_{n-1}	2 additions

We can see that we have 1 binary addition in the top row and 2 binary additions in every row beneath it. We have $n - 1$ rows with 2 binary additions. So, in total, we need:

$$1 + 2 \cdot (n - 1)$$

binary additions to add two n bit integers together.

In general, we will give operations counts with respect to the least expensive operation (which binary addition certainly is in this case). Since we need $1 + 2 \cdot (n - 1)$ binary additions, and $1 + 2 \cdot (n - 1) = O(n)$, we say that, in general, the cost of binary addition is $O(n)$.

Problem 2 (Big O Notation)

(a) I claimed that since we need $1 + 2 \cdot (n - 1)$ single binary additions, that the cost is $O(n)$. Using the definition, show why this is true.

(b) Determine the cost of subtraction on two n bit integers.

(c) Determine the cost of **multiplication** on two n bit integers. (Hint: Multiplication is just repeated addition, so try to break it down in that way.)

(d) Using your answer from part c, what can we suspect the cost of division is?

Extra Definitions and Results

Definition 4 (The Pigeonhole Principle)

If m objects are placed in n boxes, and $m > n$, then at least one box contains more than one object.

Definition 5 (The Function Pigeonhole Principle)

Let $f : S \to T$. If $|S| > |T|$, then there exists $x \neq y \in S$, such that:

$$f(x) = f(y)$$

Definition 6 (The Empty Set)

The *empty set*, denoted by $\emptyset$, is the set which has no elements. One thing that should be noted is that the empty set is a subset of **every** set. So, for any set S,

$$\emptyset \subseteq S$$

Definition 7 (Set Operations)

Let S and T be sets. We can perform the following operations of these:

Operation	Symbol	Definition
Union	$\cup$	$A \cup B = \{x : x \in A \text{ or } x \in B\}$
Intersection	$\cap$	$A \cap B = \{x : x \in A \text{ and } x \in B\}$
Difference	$\setminus$	$A \setminus B = \{x : x \in A \text{ and } x \notin B\}$

Definition 8 (Disjointness)

Let S and T be sets. These sets are said to be **disjoint** if S and T have no elements in common. Symbolically:

$$S \cap T = \emptyset$$

Problems from the Lecture Material

For all questions, we need to remember that the reasoning is far more important than the answer. As you do the problems listed, try to also write down your thought process as you solve them.

Problem 3 (Set Theory)

(a) Let $S = \{a, b, c\}$. List out all of the subsets of S. How many are there?

(b) Let T be a set with n distinct elements. How many distinct subsets of T are there?

(c) (*Extra Hard*) In a set of 10 distinct integers between 1 and 100, there exists two disjoint non-empty sets with the same sum.

Problem 4 (Quantifiers)

Determine which of the following statements are true or false. If the statement is true, justify your reasoning. If the statement is false, give a concrete example of when it fails.

(a) $\forall x \in \mathbb{Z}, \exists y \in \mathbb{Z}, xy = 0.$

(b) $\exists y \in \mathbb{Z}, \forall x \in \mathbb{Z}, xy = 0.$

(c) $\forall x > 0 \in \mathbb{R}, \exists y > 0 \in \mathbb{R}, y < x.$

(d) $\exists y > 0 \in \mathbb{R}, \forall x > 0 \in \mathbb{R}, y < x.$

(e) $\forall n \in \mathbb{N}, \exists m \in \mathbb{N}, m > n^2.$

(f) $\exists m \in \mathbb{N}, \forall n \in \mathbb{N}, m > n^2.$

Problem 5 (Functions)

Determine, for each function, whether it is injective, surjective, bijective or neither. Justify your answers.

(a) $f : \{0, 1, 2, ..., 9\} \to \{0, 1, 2, ..., 9\}$, $f(x) = 3x \mod 10$

(b) $g : \{0, 1, 2, ..., 9\} \to \{0, 1, 2, ..., 9\}$, $g(x) = 2x \mod 10$

(c) $h : \{0, 1, 2, ..., 6\} \to \{0, 1, 2, ..., 6\}$, $h(x) = 5^x \mod 7$

Problem 6 (Modular Arithmetic)

Find the solution to the following modular equations. If it does not have a solution, justify why.

(a) $3 \cdot 5 \cdot 7 \mod 9$

(b) $5 \cdot x \equiv 1 \mod 13$

(c) $3 \cdot x \equiv 1 \mod 15$

(d) $x^2 \equiv 1 \mod 7$